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الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

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Information Technology

IT438

Communication Technology

Kamal Hamza, PhD

khamza@eelu.edu.eg

Acknowledgement: *This presentation contains some figures and text from Data Communications and Networks book by W. Stallings*

Analog Signal

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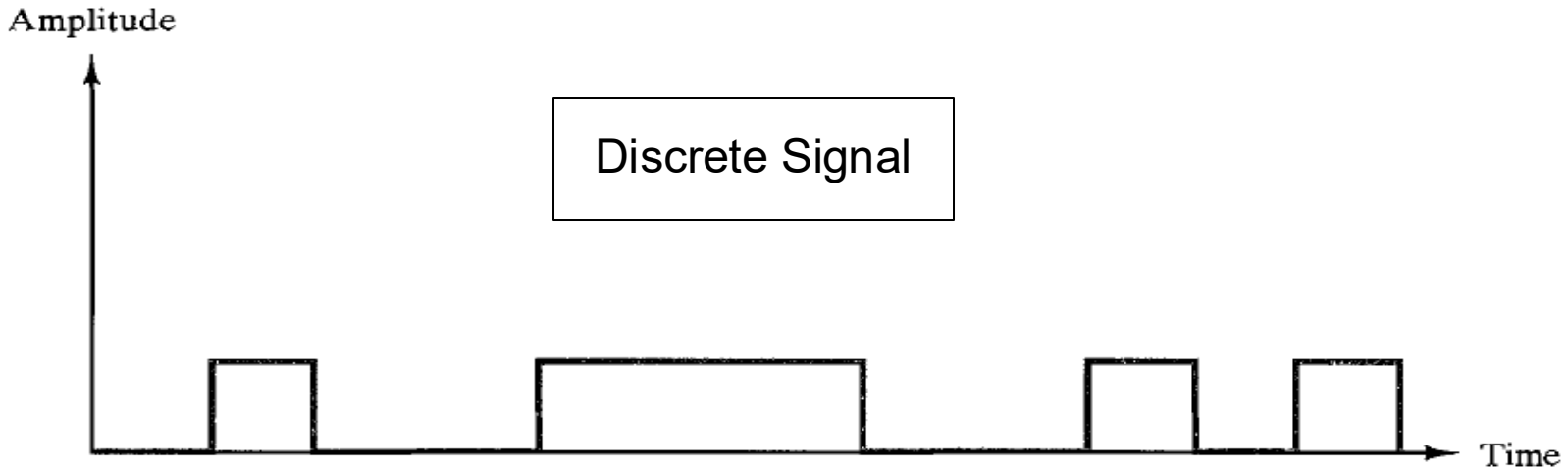
Signal Representation in the Time Domain

- Viewed as a function of time, an electromagnetic signal can be either continuous or discrete.
  - A continuous signal is one in which the signal intensity varies in a smooth fashion over time.



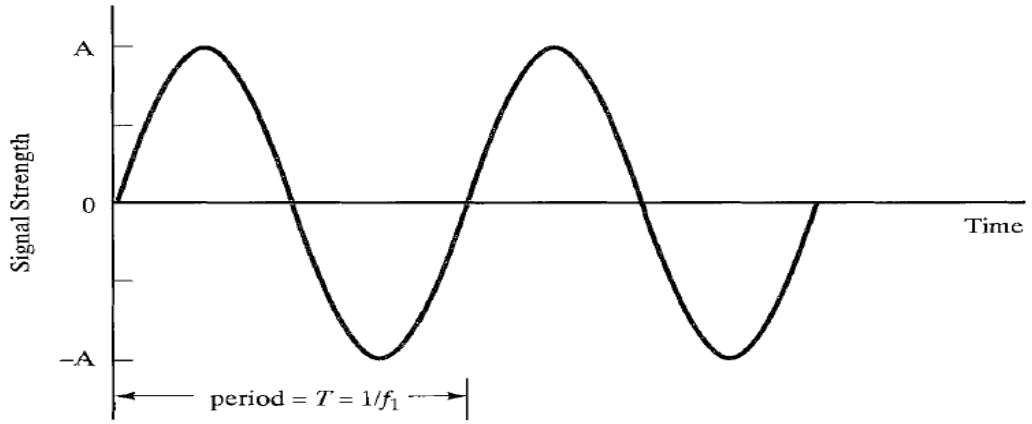
Signal Representation in the Time Domain (cont.)

- A discrete signal is one in which the signal intensity maintains a constant level for some period of time and then changes to another constant level.



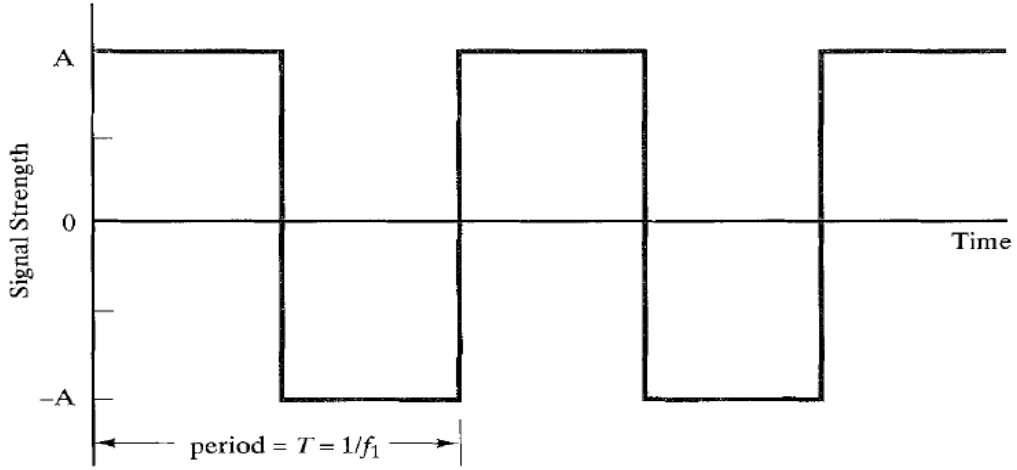
Signal Representation in the Time Domain (cont.)

Sine Wave



- Smoothly rises and falls, following a continuous curve.
- used to represent **analog signals** in communication systems.

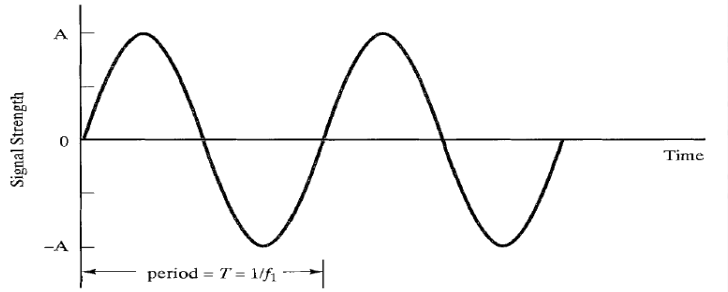
Square Wave



- Quickly switches between high and low values, creating sharp, rectangular steps.
- used in **digital systems** to represent binary data (0s and 1s).

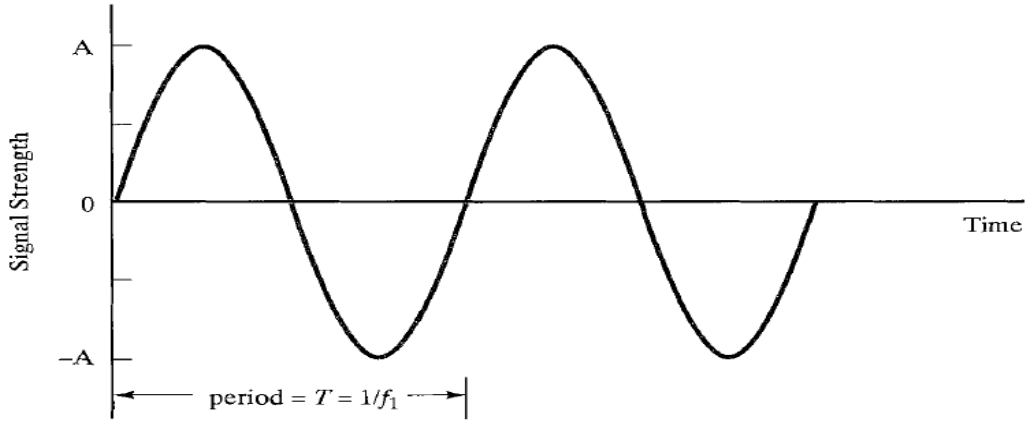
Signal Representation in the Time Domain

- A sine wave is a fundamental mathematical function that describes a smooth, periodic oscillation. Key Features of a Sine Wave:
- Amplitude (A):
  - This is the peak value (height) of the wave, representing the maximum displacement of the wave from its central position. Higher amplitude means a stronger or louder.
- Frequency (f):
  - Frequency refers to how many cycles (complete waves) occur per second, measured in Hertz (Hz). A higher frequency means the wave oscillates faster.



Signal Representation in the Time Domain

- Period (T):
  - The period is the time it takes to complete one full cycle of the wave. It is the inverse of frequency (T = 1/f). A shorter period means a higher frequency.
- Phase (ϕ):
  - Phase refers to the horizontal shift of the wave. If two sine waves have different phases, one wave may start at a different point in its cycle compared to the other. Phase differences are crucial in signal modulation and communication.

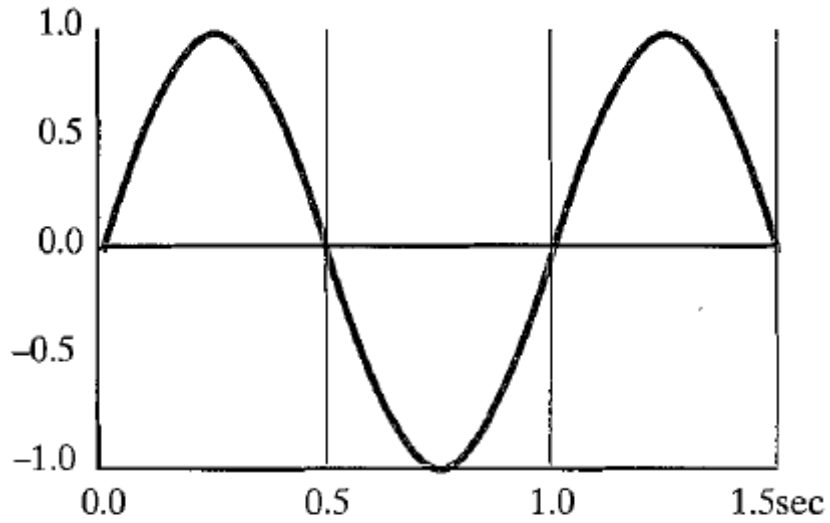


Signal Representation in the Time Domain (cont.)

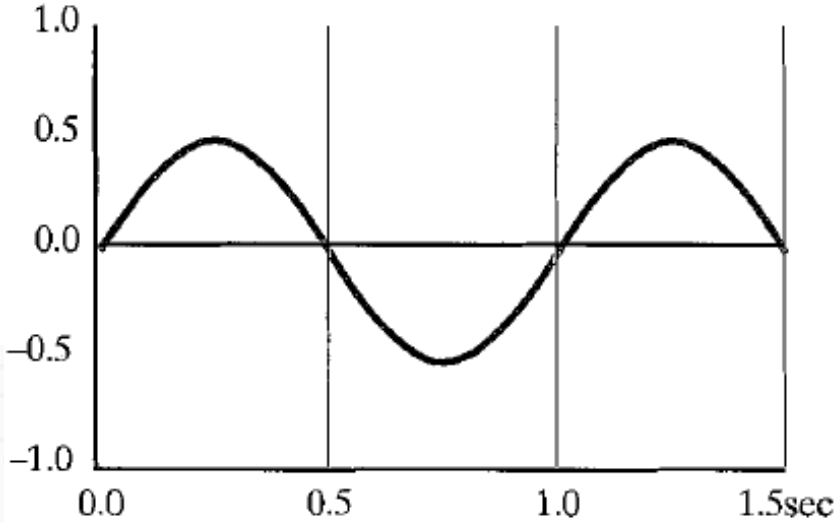
Sine Wave

$$s(t) = A \sin(2\pi ft + \phi)$$

Amplitude, Frequency, Phase



(a) $A = 1, f = 1, \phi = 0$

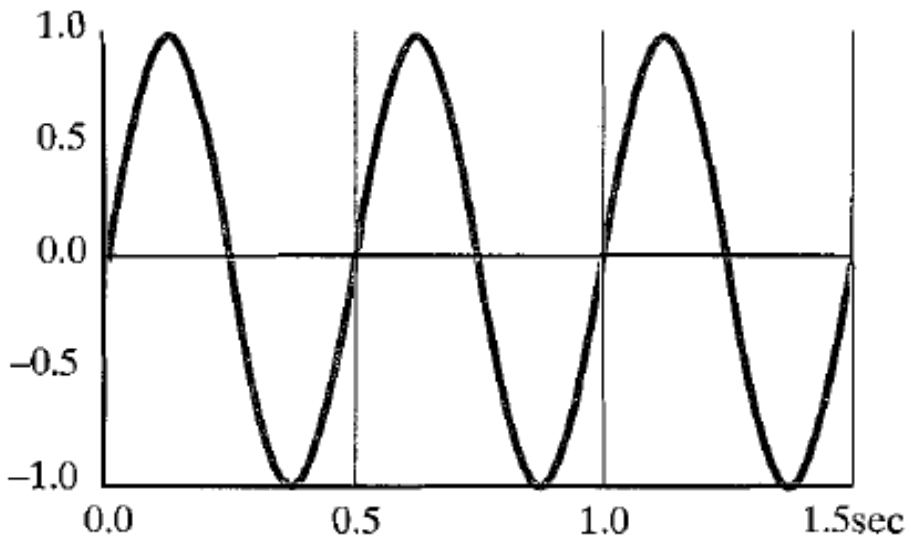


(b) $A = 0.5, f = 1, \phi = 0$

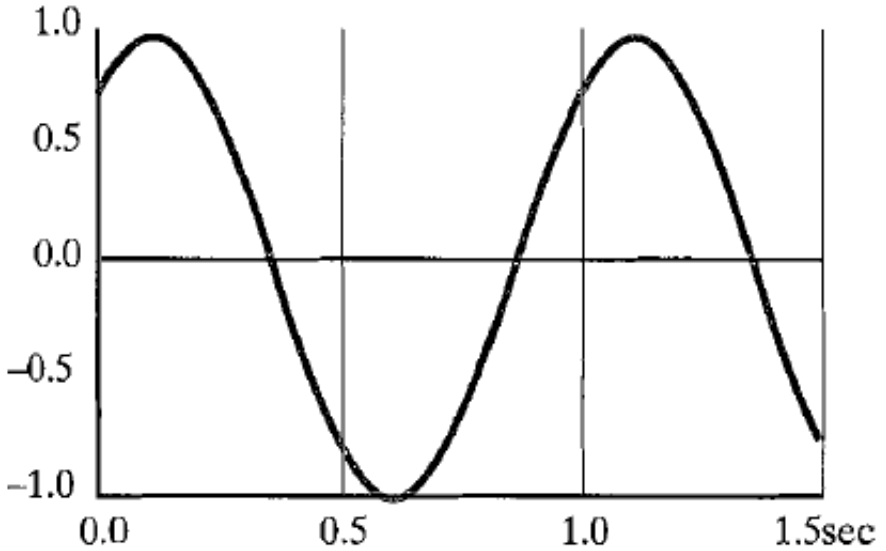
Signal Representation in the Time Domain (cont.)

Sine Wave (cont.)

$$s(t) = A \sin(2\pi ft + \phi)$$



(c) $A = 1, f = 2, \phi = 0$



(d) $A = 1, f = 1, \phi = \pi/4$

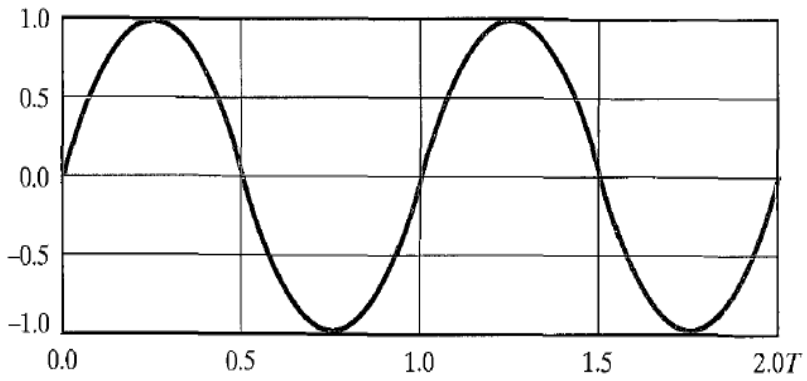
Signal Representation in the Frequency Domain

- In practice, an electromagnetic signal will be made up of many frequencies.
- For example, the signal:

$$s(t) = \sin(2\pi f_1 t) + \frac{1}{3}\sin(2\pi(3f_1)t)$$

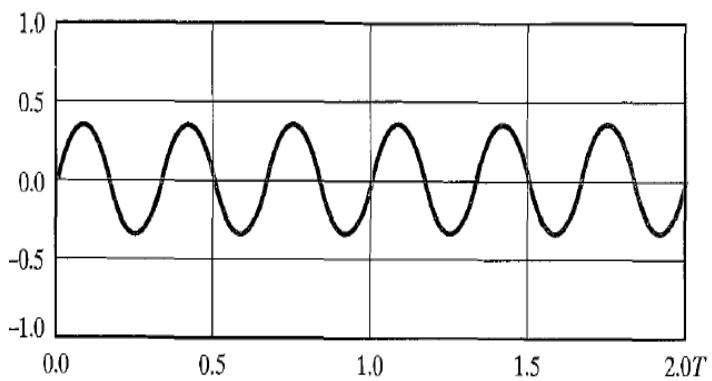
- is made up sine waves of frequencies f1 and 3f1
- The spectrum of a signal is the range of frequencies that it contains.

Signal Representation in the Frequency Domain (cont.)



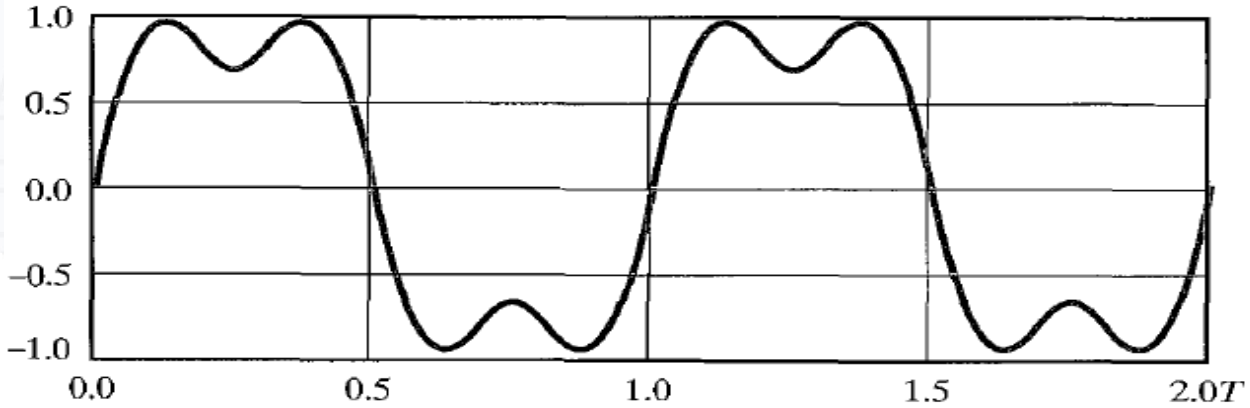
(a) $\sin(2\pi f_1 t)$

+



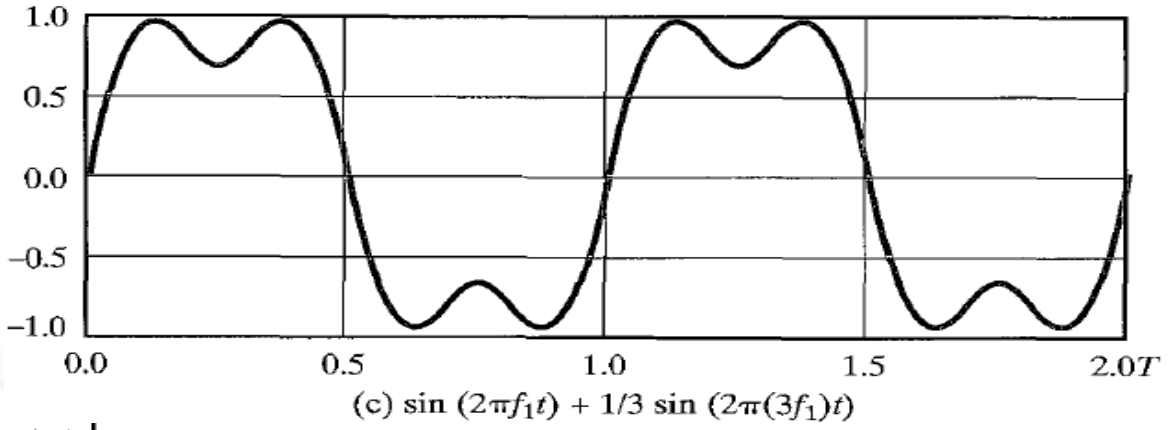
(b) $1/3 \sin(2\pi(3f_1)t)$

=



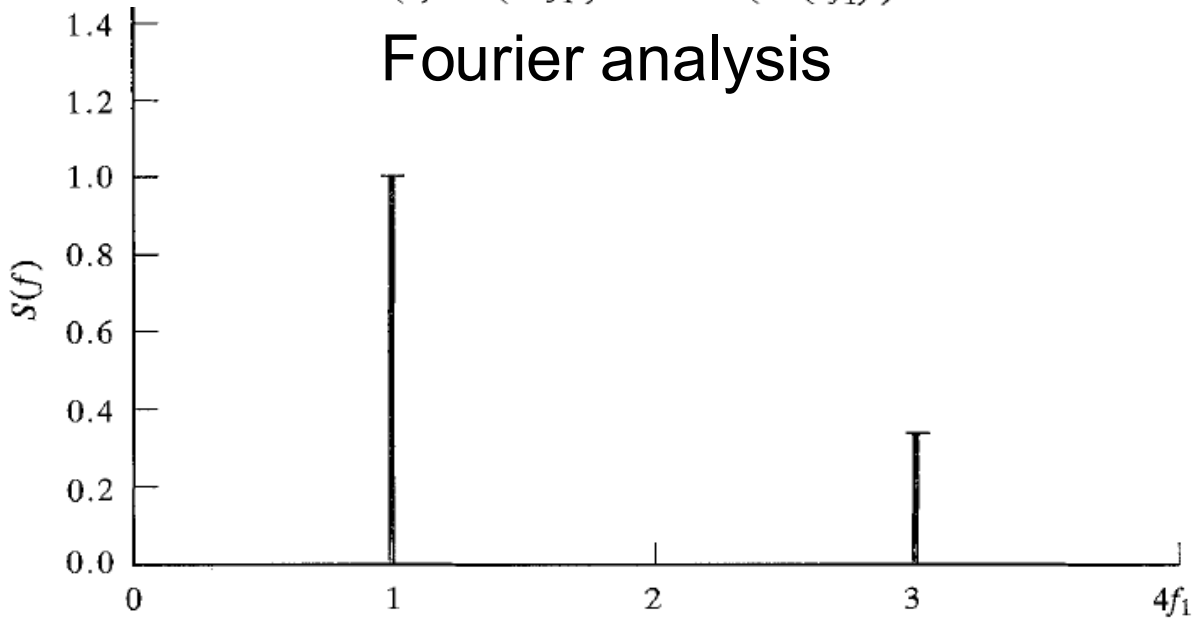
(c) $\sin(2\pi f_1 t) + 1/3 \sin(2\pi(3f_1)t)$

Signal Representation in the Frequency Domain (cont.)



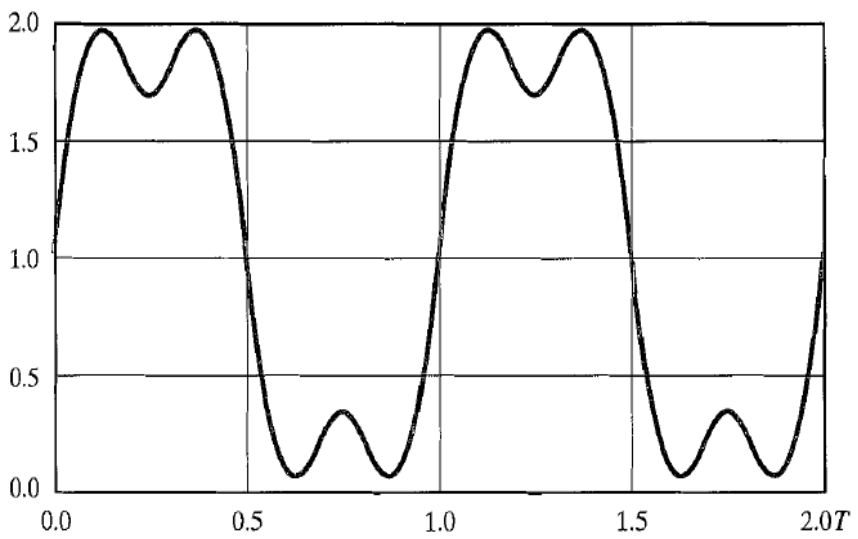
(c) $\sin(2\pi f_1 t) + 1/3 \sin(2\pi(3f_1)t)$

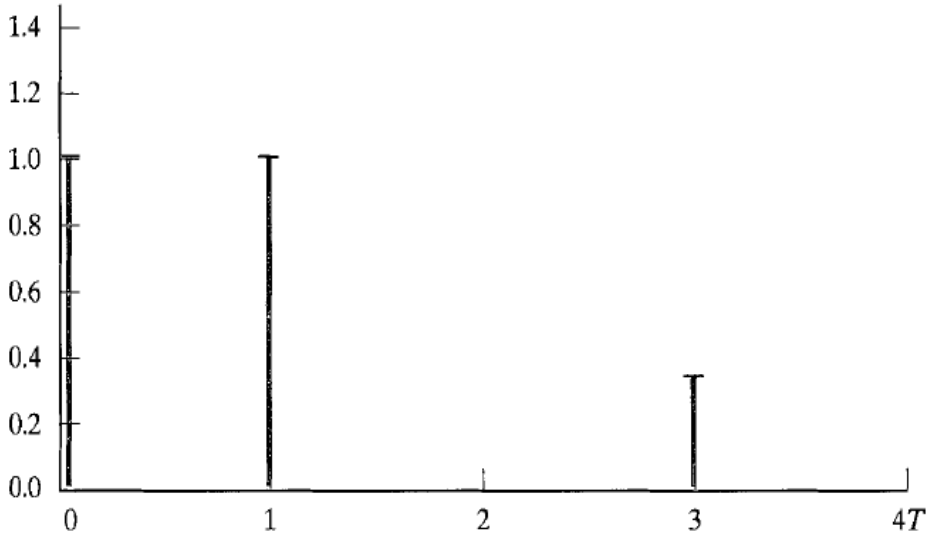
Fourier analysis



Signal Representation in the Frequency Domain (cont.)

- If a signal includes a component of zero frequency, that component is a direct current (dc) or constant component.





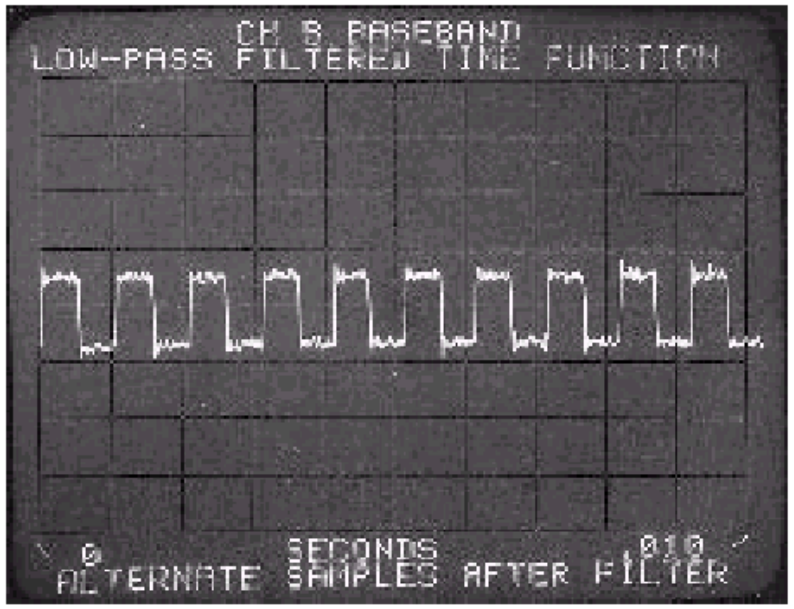
$s(t) = 1 + \sin + 1/3 \sin (2\pi(3f_1)t)$

$S(f)$

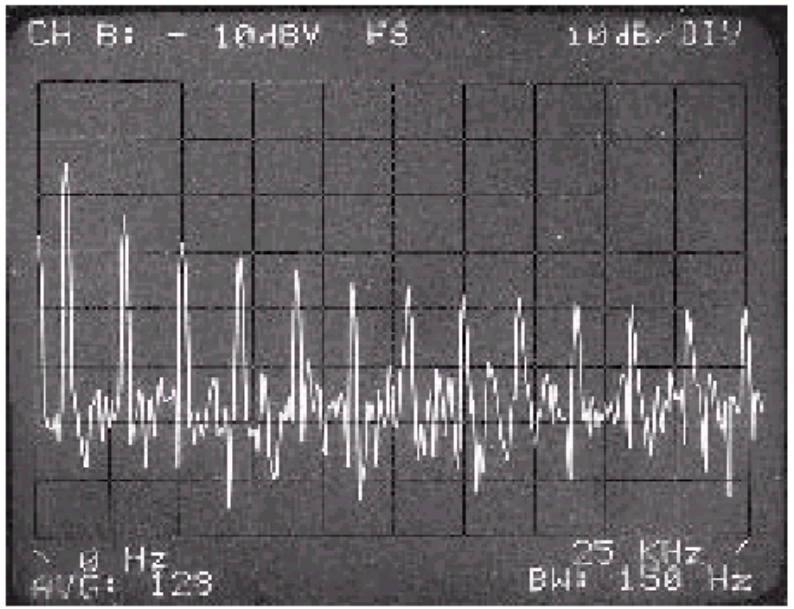
Noise and Interference

- In practical communication systems signals are blurred by noise and interference:

•Time domain

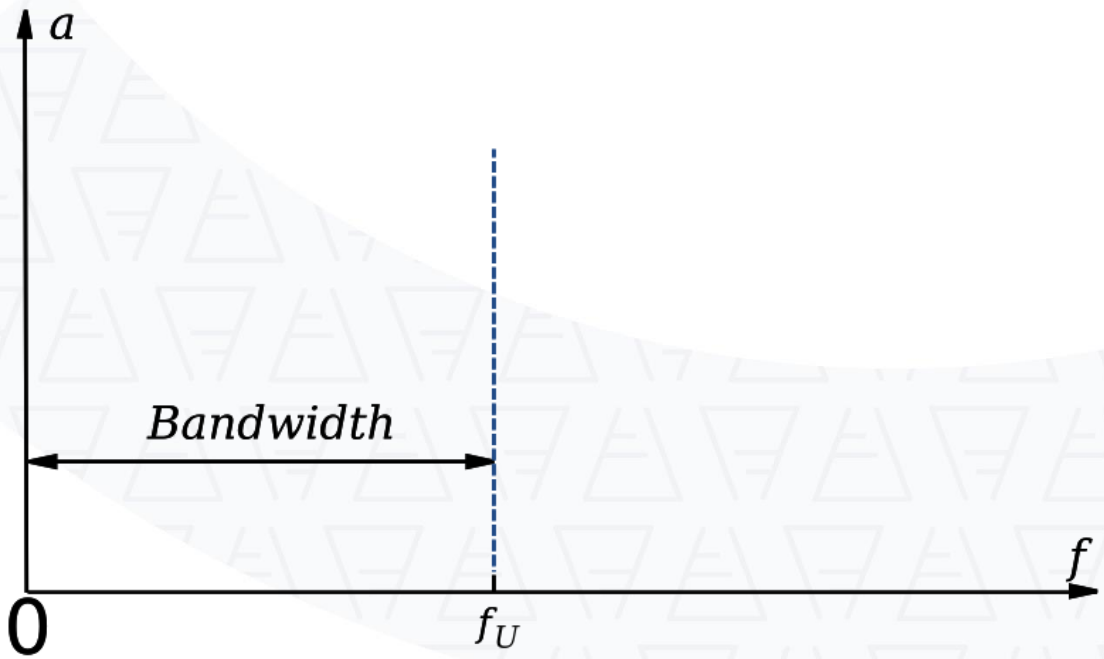


•Frequency domain



Signal Bandwidth (cont.)

Bandwidth is the difference between the upper and lower frequencies in a continuous band of frequencies. It is typically measured in unit of hertz (symbol Hz).



Signal Bandwidth (cont.)

Bandwidth in Different Communication Systems:

1. Telecommunications (Audio):

In traditional telephone systems, the bandwidth is typically limited to 300 Hz to 3400 Hz, which is sufficient for transmitting the human voice. This 3.1 kHz bandwidth filters out both very low and very high frequencies.

2. Radio Broadcasts:

AM radio stations use bandwidths around 10 kHz per channel, while FM radio uses a much larger bandwidth (around 200 kHz) to transmit better quality sound.

Signal Bandwidth (cont.)

Bandwidth in Different Communication Systems (Cont.):

3. Video and TV Transmission:

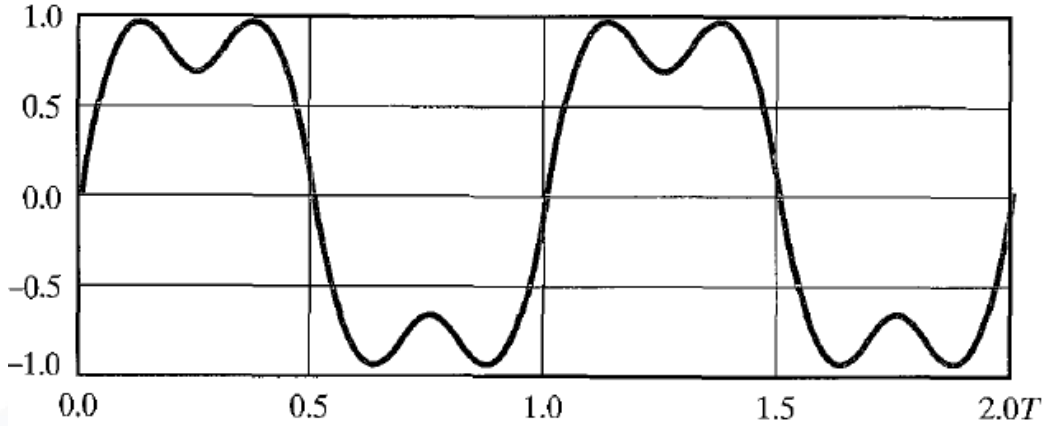
Analog TV signals use bandwidths around 6 MHz. Digital video signals (e.g., HD video) use even more bandwidth, which is why they rely on advanced compression techniques.

4. Wi-Fi and 5G:

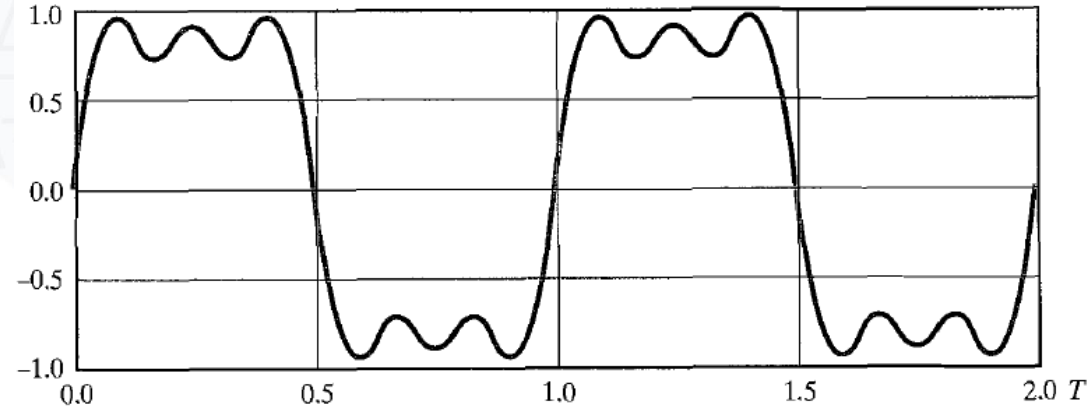
a) Wi-Fi operates in the 2.4 GHz or 5 GHz frequency bands, with bandwidths typically ranging from 20 MHz to 160 MHz. Higher bandwidth means faster internet speeds.

b) 5G technology operates at frequencies between 3 GHz and 100 GHz, with very large bandwidths available (up to several GHz), allowing it to carry massive amounts of data.

Signal Bandwidth (cont.)

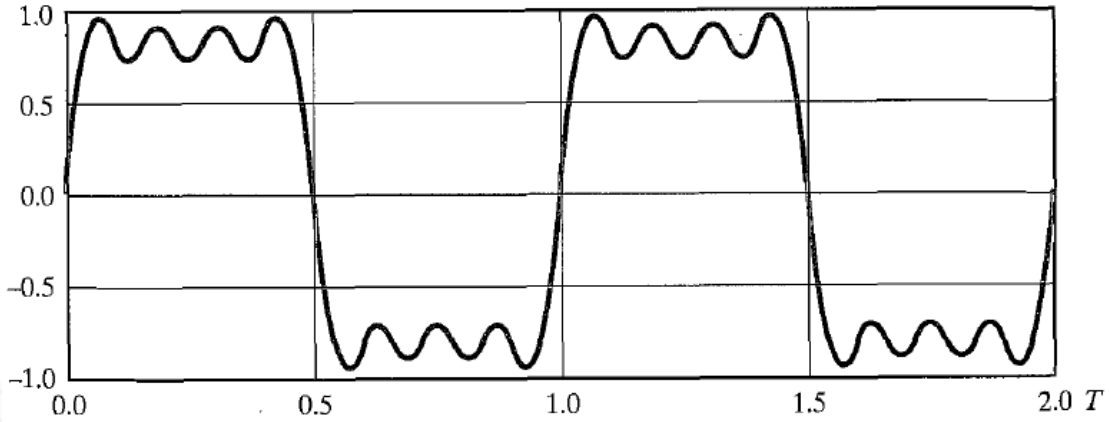


$$\sin(2\pi f_1 t) + 1/3 \sin(2\pi(3f_1)t)$$

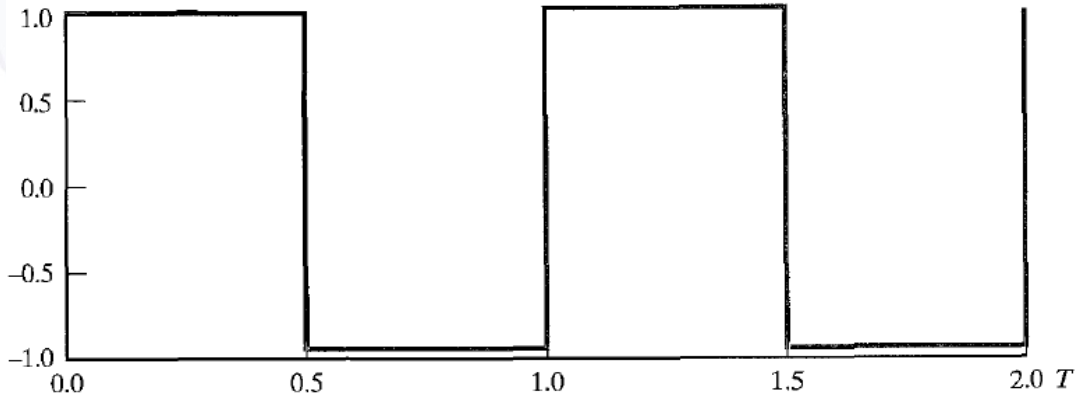


$$\sin(2\pi f_1 t) + 1/3 \sin(2\pi(3f_1)t) + 1/5 \sin(2\pi(5f_1)t)$$

Signal Bandwidth (cont.)



$$\sin(2\pi f_1 t) + 1/3 \sin(2\pi(3f_1)t) + 1/5 \sin(2\pi(5f_1)t) + 1/7 \sin(2\pi(7f_1)t)$$



$$s(t) = A \times \sum_{k \text{ odd}, k=1}^{\infty} \frac{1}{k} \sin(2\pi k f_1 t)$$



THANK YOU

QUESTIONS?





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